UMTC	COURSE WRITTEN ASSESSMENT	Course Code	EL306-1
		Version No.	7.00
		Effective Date	23 Feb 2024

Course Title: ELECTRO-TECHNICAL OFFICER **Set:** ____________

Name of Trainee: ANSWER KEY - REFERENCE **Score:** ____________

Rank: ____________ **Company/Batch:** ____________

Name of Assessor: ____________ **Date of Assessment:** ____________

Part I. **Matching Type Questions.** The following are important things that Electro-Technical Officers onboard the ship should know for the safety of the crew, the equipment, and the environment. Identify the following by matching them with the selections given below. Write your answers in the space provided before the number.

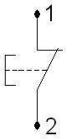
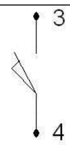
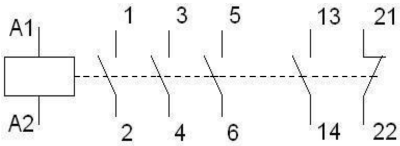
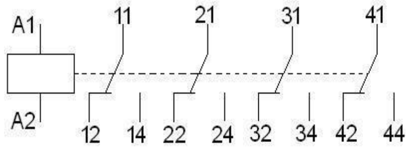
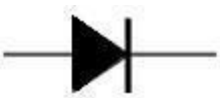
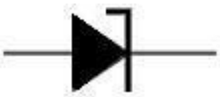
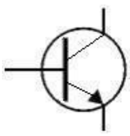
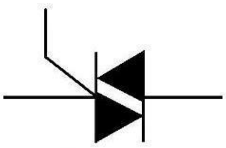
IMO	**Preferential Trip Test**	**Insulation Testing Report**	**Risk Assessment**
SMCP	**Reverse Power Test**	**Handover Report**	**Oil Spill Dispersant**
GMP	**Blackout Test**	**Incident Report**	**Lock-Out, Tag-Out**
MARPOL	**Battery Test**	**Engine Inspection Report**	**Multi-meter**

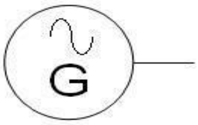
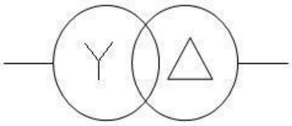
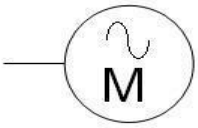
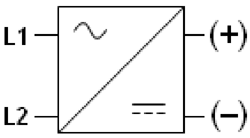
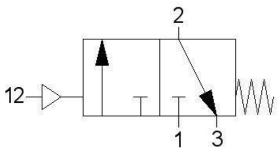
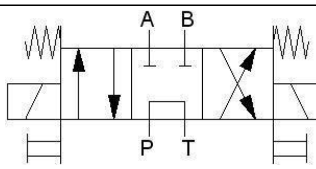
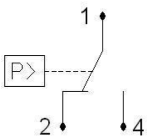
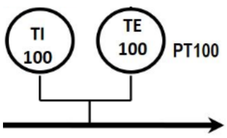
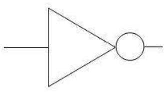
(Topic No. 1) **SMCP**	1.	A set of key phrases based on the English language, developed to cover the most important safety-related fields of verbal shore-to-ship (and vice versa), ship-to-ship and on-board communications.
(Topic No. 49) **Preferential Trip Test**	2.	A ship system that checks the integrity of the electrical arrangement on the vessel that is designed to disconnect the non-essential circuit from the main bus bar in case of partial failure or overload of the main supply.
(Topic No. 2) **Handover Report**	3.	A document created by crew members who are about to leave for disembarkation to assist their reliever to carry out their duties properly and safely.
(Topic No. 23) **Risk Assessment**	4.	A ship system that is used to identify hazards and its corresponding controls to ensure safety while performing the job.
(Topic No. 4) **Oil Spill Dispersant**	5.	A chemical agent that is sprayed on the surface of oil slick to help break it up into very small droplets, which will dilute throughout the water.
(Topic No. 5) **GMP**	6.	A guideline that comprises of a written procedure for collecting, storing, processing, and disposing of garbage generated onboard ship as per regulations provided in Annex V of MARPOL.
(Topic No. 23) **Lock-Out, Tag-Out**	7.	This ship equipment is a disconnection and isolation tool that is capable of warning crew members of a hazard and controlling the risks from the hazard during maintenance or repair work.
(Topic No. 50) **Incident Report**	8.	A tool that documents any event that may or may not have caused injuries to a person or damage to a company asset to capture injuries and accidents, near misses, property and equipment damage, health and safety issues, security breaches and workplace misconduct.
(Topic No. 49) **Blackout Test**	9.	A ship system that verifies capability of the vessel to operate on backup or emergency power in case a power outage happens.
(Topic No. 3) **MARPOL**	10.	A convention that includes regulations aimed at preventing and minimizing pollution from ships - both accidental pollution and that from routine operations - and currently includes six technical Annexes.

Part II. **Identification.** Identify the following terminologies used in electro-technical equipment relevant to the job of an Electro-Technical Officer onboard the ship. Write your answer in the space provided before the number.

(Topic No. 41) **Diesel Engine**	11.	A type of marine propulsion equipment that converts mechanical energy from thermal forces – designed as a reciprocating internal combustion to produce mechanical power.
(Topic No. 41) **JCWS**	12.	A ship closed-circuit water system passing from the engine that returns through a cooler to the circulating pump and then to the engine under controlled condition where an expansion tank is placed at a sufficient height to allow the venting and water make-up in the system.
(Topic No. 41) **Turbocharger**	13.	An engine-room equipment integral part of the ship's main engine that uses the exhaust gases in order to increase the overall efficiency of the engine by compressing air to increase its density for combustion
(Topic No. 28) **Control Valve**	14.	A process control equipment that can be found on engine room machinery or system that regulates, directs, or controls the flow of a fluid by opening, closing, or partially obstructing various passageways.
(Topic No. 28) **Viscometer/Viscosimeter**	15.	A ship process control system used to measure, control, and regulate the viscosity of fuel oil at the engine inlet consistings of a capillary tube inside of which a gear pump is fitted that rotates at very slow speed.
(Topic No. 32) **Capacitive Sensor**	16.	A sensor module used in control systems that are contactless types capable of detecting proximity of liquid or solid target by emitting electrical field
(Topic No. 32) **Reed Switches**	17.	A sensor module used in control systems that are contactless types that are capable of detecting of open or close position actuators by detecting the presence of magnetic fields.
(Topic No. 42) **Purifier**	18.	An engine-room machinery that is used to separate two liquid substances based on their difference in density with the use of gravity disc to separate solid and water and sealing water as an operator with two-holed lower discs – one for water and other for oil. (wartsila)
(Topic No. 42) **Stern Tube**	19.	A hollow tube that accommodates the bearings, the seal boxes and the propeller shaft of a ship filled with oil, grease or water and forms a barrier between the water outside of the ship and the engine room.
(Topic No. 42) **OWS**	20.	A shipboard equipment used to separate engine room bilge oil and water mixtures into their separate components, specific to bring down the oil content of the water that has to be discharged overboard.
(Topic No. 42) **Boiler**	21.	An engine-room pressure vessel used on ships in which the water is heated to evaporate and generate steam for various consumers on board.
(Topic No. 42) **Air Compressor**	22.	One of the functions of the air receiver tank in the compressed-air generating plant onboard the ship is to automate which engine-room machinery? (wartsila)
(Topic No. 42) **Sewage Treatment Plant**	23.	A ship facility that uses a multistage process to reduce or remove the pollutants and kill disease-causing organisms from wastewater, sewage before it is discharged into the sea.
(Topic No. 51) **HV Step-down Transformer (HVSC)**	24.	A ship equipment that allows above 1,000 volts power supply directly from the wharf to be used by the vessel to ensure operation of the machinery systems and installations onboard while berthed and engines are shut-off to reduce fuel consumptions and carbon emissions at port. (for revision)
(Topic No. 52) **HV – SF6 CB**	25.	A ship electrical power distribution equipment that switches and protects the connected equipment that uses a gas called sulfur hexafluoride to quench the occurrence of electric arcs on operation of power systems rated above 1,000 volts. (for revision)

Part III. **Identifying Electro-Technical Symbols**. Identify the given symbols below and briefly describe their function or operation. Write your answers in the columns provided.

Component Symbol	Name / Description	Operation / Function
26. 	*(Topic No. 18)* **Normally-open contact switch**	Triggered when there is an external force acting upon it to let electricity flow
27. 	*(Topic No. 18)* **Normally-closed contact switch**	Triggered when there is an external force acting upon it to cut off electricity flow
28. 	*(Topic No. 16)* **Normally-closed Pushbutton switch**	Controls devices or machine by pressing it
29. 	*(Topic No. 16)* **Normally-open Limit switch**	Detects the presence or absence of an object and triggers when predetermined point is reached
30. 	*(Topic No. 16)* **Magnetic Contactor Three-phase**	A switching device that control high power devices
31. 	*(Topic No. 16)* **Control Relay**	A switching device that control low power devices
32. 	*(Topic No. 17)* **Diode**	Lets current flow in one direction only
33. 	*(Topic No. 17)* **Zener Diode**	Lets current flow in one direction and reverse
34. 	*(Topic No. 17)* **NPN - Transistor**	Use for amplifying or switching electrical signals
35. 	*(Topic No. 17)* **TRIAC**	Acts as a switch allowing control of ac power

<table>
<tr><th>Component Symbol</th><th>Name / Description</th><th>Operation / Function</th></tr>
<tr><td>36.
</td><td>(Topic No. 48)
Generator</td><td>Converts mechanical to electrical energy</td></tr>
<tr><td>37.
</td><td>(Topic No. 48)
Transformer</td><td>Transfers electrical energy to one or more circuit by either increasing or decreasing it</td></tr>
<tr><td>38.
</td><td>(Topic No. 48)
AC Motor</td><td>Converts electrical to mechanical energy</td></tr>
<tr><td>39.
</td><td>(Topic No. 48)
Rectifier Module</td><td>Converts AC to DC power</td></tr>
<tr><td>40.
</td><td>(Topic No. 24)
3/2-way Pilot-Actuated Valve with Spring Reset</td><td>Can be triggered either pneumatically or mechanically.</td></tr>
<tr><td>41.
</td><td>(Topic No. 26)
4/3-way Solenoid-Actuated Valve</td><td>Can be triggered electrically because of solenoid</td></tr>
<tr><td>42.
</td><td>(Topic No. 30)
Pressure Switch</td><td>Monitors the pressure of fluids</td></tr>
<tr><td>43.
</td><td>(Topic No. 30)
Temperature Detectors</td><td>Monitors temperature of equipment or machines</td></tr>
<tr><td>44.
</td><td>(Topic No. 32)
NOT Logic Gate / Inverter Circuit</td><td><table><tr><th>A</th><th>Q</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table></td></tr>
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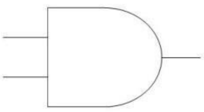</td><td>(Topic No. 47)
AND Logic Gate</td><td><table><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table></td></tr>
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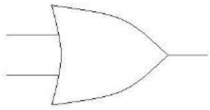
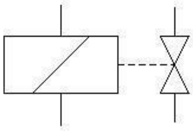
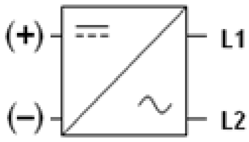
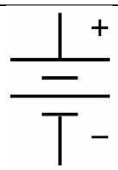
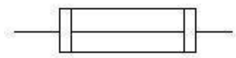
<tr><td>46. </td><td>(Topic No. 47)

OR Logic Gate</td><td><table><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table></td></tr>
<tr><td>47. </td><td>(Topic No. 42)

Control Valve,
Solenoid-Actuated</td><td>Electrically controlled valve</td></tr>
<tr><td>48. </td><td>(Topic No. 52)

Electronic DC-AC
Converter Modules</td><td>Converts dc to ac power</td></tr>
<tr><td>49. </td><td>(Topic No. 36)

Battery</td><td>Supplies dc voltage</td></tr>
<tr><td>50. </td><td>(Topic No. 35)

Fuse</td><td>Cuts off current when there is overloading</td></tr>
</table>

Part IV. **Multiple Choice Questions.** Identify what is required by encircling the letter for the answer of your choice. (for review and revision in question form)

(Topic No. 13)

51. What is a measuring instrument that is used to measure current without actually making any connection in the wire being measured?
    [✓ a. Clamp meter]
    b. Megger tester
    c. Multimeter
    d. Voltage detector

52. When using an analog/manual multimeter voltage, current, resistance in a circuit, always start with __________________.
    a. the mid-range of the meter
    b. the lowest range of the meter
    [✓ c. the highest range of the meter]
    d. any range of the selector in the meter

53. Which of the following best describes the results of an insulation resistance testing conducted on low voltage electrical motors?
    a. Typical values should be in micro-ohms.
    b. Typical values should be in milli-ohms.
    c. Typical values should be in kilo-ohms.
    [✓ d Typical values should be in mega-ohms.]

54. What is a measuring instrument that is used to measure voltage without actually making any connection in the wire being measured? (for revision of choices)
    a. Clamp meter
    b. Contact voltage detector

c. Multimeter
d. Non-contact voltage detector

55. Electrical power is measured in which of the following electrical units?
a. Amperes
b. Watts
c. Ohms
d. Volts

56. Which of the following electrical testing procedures is best suitable in determining continuity of a control circuit while the circuit is still energized? (for revision of question)
a. Resistance testing
b. Current testing
c. Voltage testing
d. Insulation testing

(Topic No. 15)
57. What is the process of joining metal parts of electronic components to form a mechanical or electrical bond?
a. Breadboarding
b. Etching
c. Soldering
d. Welding

58. What is the process in producing electrical prints on boards in creating circuit plates?
a. Breadboarding
b. Etching
c. Soldering
d. Welding

(Topic No. 21)
59. Among the following things that any personnel should be familiar with for electrical-related emergencies onboard ships, what is supposed to be identified has priority? (for revision)
a. Location of emergency shut-off switches
b. Location of first-aid supplies
c. Location of the safety hook
d. Location of a telephone

60. If an electric shock victim is still in contact with an energized circuit and activating the emergency shut-off switch is not yet possible, what should you do?
a. Grab the victim with one hand and then pull him away from the equipment.
b. Pry the victim loose using a dry wood pol.
c. Throw a rope around the victim and pull.
d. Pry the victim loose with a metal pol.

(Topic No. 22)
61. What is the mechanical workshop process that allows the smoothening of the finish of the metal parts used for shipboard machinery or equipment? (for checking)
a. Flaring
b. Grinding
c. Soldering
d. Welding

62. Which of the following hand tools is used for gripping, fastening, turning, tightening and loosening pipes, pipe fittings, nuts and bolts?
a. Wrenches
b. Files
c. Screw drivers
d. Pliers

63. What is the mechanical workshop process that allows the connection of copper tubes to each other or another kind of fitting by creating an approximately conical shape at one end?
    a. Flaring
    b. Grinding
    c. Soldering
    d. Welding

64. What do you call the devise consisting of two parallel jaws for holding a workpiece of which one of the jaws is fixed and the other movable by a screw, a lever, or a cam?
    a. Vernier caliper
    b. Chain block
    c. Bench vise
    d. Vise grip

65. What do you call the workshop equipment used to raise or lower heavy loads with relative ease using internal gears that wind the chains around, lifting the object that is attached to the end or hook? (for revision)
    a. Vernier caliper
    b. Chain block
    c. Bench vise
    d. Vise grip

(Topic No. 20)

66. This shipboard system ensures that all maintenance is carried out at adequate interval, under the intended schedule while ensuring efficiency and continues operability.
    a. Permit to work system
    b. Automation system
    c. Planned maintenance system
    d. Instrumentation system

(Topic No. 16)

67. Which of the following describes the operation of a step-down transformer?
    a. Decrease the output power
    b. Increase the output power
    c. Decrease the output voltage
    d. Increase the output voltage

68. How would you describe a circuit which contains two resistors with un-equal resistances in parallel?
    a. Large current flows in the larger resistor
    b. Smaller resistance has smaller conductance
    c. Potential difference across each is the same
    d. Same amount of current flows in the two resistors

69. Which of the following bulbs will have the least resistance?
    a. 220V, 60W
    b. 220V, 100W
    c. 115V, 60W
    d. 115V, 100W

70. In analyzing AC electrical circuits, the term impedance refers to which of the following? (to revise the choices)
    a. The practical effect of reactance in a circuit
    b. The practical effect of reactance in a circuit
    c. The practical effect of inductance for a single group of components
    d. The practical effect of resistance, capacitance and inductance in a circuit

71. Which of the following describe how a capacitor works in an electrical/electronic circuit?
    a. It converts electricity into magnetism.
    b. It transforms electrical energy into heat energy.
    c. It stores electrical energy like a fast-charging battery.

d. It increases a low voltage electrical signal into a higher voltage electrical signal.

72. Which one of the following describe why most circuits onboard ships are supplied by AC electricity rather than DC electricity? (to review the choices)
    a. The waveform of an AC signal makes electric motors run smoother than with a DC signal.
    b. The voltage level of AC electrical signals is easily changed.
    c. 100 volts of AC is safer than 100 volts of DC.
    d. DC generators are not available on ships.

(Topic No. 17)

73. Which of the following electronic modules refer to the circuits that prepare sensor output signals into a format that may be understood by controllers?
    a. Amplifiers
    b. Converters
    c. Transmitters
    d. Rectifiers

74. Which one of the following describe the operation of a SCR in electronic circuits?
    a. It can be triggered on positive gate voltages.
    b. It can be triggered on negative gate voltages.
    c. It can be triggered with either a positive or a negative gate voltage.
    d. It can be triggered with neither a positive nor a negative gate voltage.

75. Zener diodes are used in electronic circuits as __________.
    a. reference resistances
    b. reference current elements
    c. reference voltage elements
    d. reference phase angle elements

76. Which of the following is not a possible function of a semiconductor diode in electronic circuits?
    a. Signal regulator
    b. Signal indicator
    c. Signal detector
    d. Signal filter

77. Which of the following terms refer to the temperature sensing elements that are made out of pure metal capable of translating temperature changes with resistance changes?
    a. Thermistor
    b. Thermo-electric
    c. PT100
    d. Piezo-electric

78. A ceramic capacitor marked "104J" indicates a capacitance of ___________.
    a. 0.1 µF ±5%
    b. 100 nF ±10%
    c. 1,000 nF ±5%
    d. 100,000 pF ±10%

(Topic No. 19)

79. Which of the following motor control components delays the changing of contact positions whenever the coupled coils are energized or de-energized?
    a. Magnetic contactors
    b. Circuit breakers
    c. Electrical timers
    d. Overload relays

80. The speed of rotation of stepper motors are determined with _________. (for review)
    a. the load applied to the motor
    b. the current supplied to the motor
    c. the magnitude of the voltage applied

d. the pulse of electricity

81. How many percent of voltage increase or decrease that the motor may be able to carry with its rated horsepower?
    a. ± 5% of the rated voltage
    b. ± 10% of the rated voltage
    c. ± 15% of the rated voltage
    d. ± 20% of the rated voltage

82. Which of the following motor control circuit types do not implement a reduced-voltage starting technique?
    a. Wye-to-Delta Starters
    b. Electronic Soft Starters
    c. Across-the-Line Starters
    d. Auto-Transformer Starters

83. What is the current flowing through the line when the motor is operating at rated power, rated frequency and rated voltage?
    a. Inrush current
    b. Full-load current
    c. No-Load current
    d. Rotor-blocked current

(Topic No. 23)

84. Which of the following best describes the effect of over voltage when applied to an electrical conductor?
    a. It will be indicated by a color change in the insulation.
    b. It will cause thermosetting insulation to go hard.
    c. It will cause static electrical charge build up.
    d. It will cause insulation breakdown.

85. Which one of the following describe the meaning of “voltage stress” when applied to ship electrical cables?
    a. Voltage per meter of cable length
    b. Damage done to insulation after a voltage was applied across it
    c. Voltage drop caused by a right angle bend with a radius of 5 times the cable diameter
    d. Voltage per millimeter of insulation

86. When a running 3-phase induction motor encounters single-phasing, _______________. (for revision of choices, better construction of question)
    a. the motor will be immediately put on a stop because of lost power
    b. the motor will continue to run, but the direction of rotation will change
    c. the motor will continue to run, but is inhibited to operate at its rated power
    d. the motor will be immediately put on a stop because of increased line currents

87. Which one of the following describes the multiplier that may be applied to the rated power that will serve as spare to the capacity of the electric motor?
    a. Duty Cycle
    b. Frame Size
    c. Power Factor
    d. Service Factor

88. The IP number indicated on electric motor specification plate indicates ____________.
    a. the protection provided by the enclosures
    b. the regulation provided by the windings
    c. the control provided by the armatures
    d. the torque provided by the rotors

89. Which of the following describes the main function of protective relay in circuit breakers?
    a. Close the contacts when the actuating quantity reaches a certain predetermined value.
    b. Limit arcing current during the operation of the circuit breaker.
    c. Provide additional safety in the operation of circuit breaker.
    d. Protect the lines from any stray voltages.

90. Which of the following circuit connection for motor control circuits allows a three-phase motor to change its direction of rotation? (improve ques and choices)
    a. Blocked current lines
    b. Single phase controlled
    c. Interchanged current lines
    d. Primary resistance controlled

91. What is the equivalent charge of two 200-Ah batteries connected in parallel?
    a. 100 Ah
    b. 200 Ah
    c. 300 Ah
    d. 400 Ah

(Topic No. 24)

92. In hydraulics, movement of actuator depends on the ___________.
    a. flow
    b. pressure
    c. tank level
    d. temperature

93. Which of the following refer to the components required in compressed air generating plants to automate control of the compressor units?
    a. Solenoid Valves
    b. Pressure Switches
    c. Pressure Transmitters
    d. Pressure Relief Valves

94. A double solenoid 5/2-way valve uses ____________ for actuation.
    a. Electrical
    b. Hydraulic
    c. Pneumatic
    d. Mechanical

95. What is the relationship between gas pressure and volume?
    a. Directly proportional at any condition
    b. Inversely proportional at constant temperature
    c. Directly proportional at constant temperature
    d. Inversely proportional at any condition

(Topic No. 25)

96. What component in the compressed-air generating plant is necessary to reduce moisture in the compressed-air signal before distributing it into the pneumatic system?
    a. Air Compressor
    b. Air Receiver
    c. Air Cooler
    d. Air Dryer

97. Which of the following in a typical compressed-air service unit delivers a metered quantity of oil mist into the air distribution system?
    a. Filter
    b. Regulator
    c. Lubricator
    d. Manometer

(Topic No. 26)

98. In a hydraulic system, reducing the oil flow will cause the torque to (for review the correct answer)
    a. fluctuate.
    b. be reduced.
    c. be increased.

d. remain the same.

99. Energy not used to move loads in hydraulic system turns into what?
a. Heat
b. Noise
c. Pressure
d. Electricity

100. What creates pressure in a hydraulic system?
a. The pump
b. The relief valve
c. The resistance to flow
d. The directional control valve

101. Which of the following best describes the function of a counterbalance valve in hydraulic systems?
a. Reduce load
b. Prevent creeping
c. Unload the delivery pump
d. Support a load and prevent creeping

(Topic No. 27)
102. Accumulators in hydraulic systems onboard ships use what type of gas for charging?
a. Oxygen
b. Nitrogen
c. Argon
d. Compressed Air

103. What is the primary purpose of pilot-operated check valves in hydraulic equipment?
a. To reduce load
b. To prevent creeping
c. To prevent overload
d. To increase pressure

(Topic No. 29)
104. Which one of the following describe the function of a thermistor?
a. Control temperature
b. Calibrate temperature
c. Detect temperature
d. Regulate temperature

105. What is the primary purpose of automation in marine process control loops? (to review choices or change the question)
a. Maintain some parameter near a desired value or within a required range.
b. Eliminate loads or disturbances in the control system.
c. Regulate the sequence of operations in a process.
d. Monitor operations associated with a process.

106. Which of the following refers to the component that provides current-proportioning control in a marine process control system? (to review question)
a. LED
b. PID
c. PLC
d. RTD

107. Which of the following is the typical instrumentation signal range for process control systems on ships?
a. 0 mV – 20 mV
b. 0 V – 20 V
c. 4 mA – 20 mA
d. 4 A – 20 A

108. Which of the following terms refer to electrical component that is capable of converting AC signals into DC? (check/revise)
   a. converters
   b. inverters
   c. rectifiers
   d. transformers

(Topic No. 31)

109. Which of the following terms refer to the temperature sensing elements that are made out of pure metal capable of translating temperature changes with resistance changes? (same as ques #77)
   a. Thermistor
   b. Thermo-electric
   c. PT100
   d. Piezo-electric

110. Which of the following describes the sensing elements that are commonly installed in local display pressure gauges because of its robust construction and simple operation, using c-shaped tube rings with oval cross-sections?
   a. Strain gauges
   b. Bourdon tubes
   c. Bellows elements
   d. Rubber diaphragms

111. Which of the following describes the output signal of temperature switches when a temperature is reached at the measurement point? (to revise choices & question)
   a. 4 A – 20 A
   b. 0 V or 24 V
   c. 0.2 bar – 1.0 bar
   d. 4 mA – 20 mA

112. Which of the following describes the function of a pressure transmitter in a control system marine control system?
   a. Sends a normalized voltage, current, or air pressure signal to the controller proportional to pressure.
   b. Sends a pressure signal to the controller proportional to a voltage or current.
   c. Sends a mechanical signal to the controller proportional to pressure.
   d. Displays a direct visual reading of the pressure.

113. Which one of the following describes the instrumentation components that are made of two different metals bonded together that when heated may cause one side to expand more than the other and the resulting bending is translated into a temperature reading by mechanical linkage to a pointer?.
   a. Thermocouple wires
   b. Diaphragm plates
   c. Bi-metallic strips
   d. Strain gauges

114. Which of the following describes the function of a controller in a closed-loop marine control system?
   a. It compares and computes the difference between the measured variable and the setpoint, corrects by modifying the setting of the final control element until the loop reaches steadystate.
   b. It eliminates loads and disturbances in a control system to make the measured variable and the setpoint value equal.
   c. It constantly re-adjusts the manipulated variable to keep it equal, or as near as possible, to the setpoint value.
   d. It decides what action to take to make the manipulated variable equal to the controlled variable.

(Topic No. 33)

115. Which of the following refers to the hardware that provides connection between the essential parts of a computer system thru electrical buses controlled with built-in integrated circuit chip sets?
   a. RAM
   b. ROM

c. Adapter cards
d. Motherboards

116. A computer system connected into a data network thru an Ethernet Hub uses its ____________ to communicate with other computers in the system.
   a. graphic-user interface
   b. input/output interface
   c. network interface
   d. sound interface

(Topic No. 34)

117. In programmable logic controls a logical _____________ operation is required for performing series connection of physical input devices.
   a. AND
   b. OR
   c. NOT
   d. NOR

118. In programmable logic controls a logical _____________ operation is required for performing parallel connection of physical input devices.
   a. AND
   b. OR
   c. NOT
   d. NOR

(Topic No. 35)

119. Which of the following program addresses are not visible on screens, front panel displays, nor schematic diagrams of programmable logic controllers because of their function and operation in the system?
   a. Input addresses
   b. Memory addresses
   c. Output addresses
   d. Application addresses

120. What are electro-mechanical actuators that when energized applies straight line of force to directional control valves to move spools, discs, or poppet valves?
   a. Control relays
   b. Pushbutton switches
   c. Solenoid valves
   d. Magnetic contactors

(Topic No. 40)

121. Modern marine refrigeration systems like air-conditioning systems and provision refrigerators are based on the principle of _________________________________.
   a. rapid expansion with recirculation of refrigerant
   b. vapor compression with recirculation of refrigerant
   c. rapid expansion requiring periodic replacement of refrigerant
   d. vapor compression requiring periodic replacement of refrigerant

122. Which of the following components are found on the high-pressure side of the hotel refrigeration equipment?
   a. Evaporator
   b. Condenser
   c. Suction Service Valve
   d. Suction Line Accumulators

123. Which of the following describes the function of the thermostatic expansion valve?
   a. Constant superheat of vapor refrigerant at the evaporator outlet
   b. Constant condenser temperature
   c. Constant evaporator pressure
   d. Constant flow of refrigerant

124. What is the disadvantage of a manually operated expansion valve in a refrigeration system?
   a. It cannot control refrigerant flow.
   b. It cannot be used with modern refrigerants.
   c. It is unresponsive to changes in heat loads.
   d. It is very sensitive to changes in load making adjustment very hard.

125. If a running refrigeration system regularly requires charge of refrigerant, which of the following should be done?
   a. Replace the lost refrigerant with nitrogen.
   b. Find the leak, rectify the problem and then charge the system.
   c. Charge the system, find the leak, rectify the problem and then charge the system again.
   d. Charge the system periodically, since modern refrigerants are cheap, non-toxic & environment-friendly.

126. Which of the following describes the function of an automatic expansion valve?
   a. Constant condenser pressure and temperature
   b. Constant evaporator temperature
   c. Constant evaporator pressure
   d. Constant flow of refrigerant

127. Which of the following describes an absolute pressure?
   a. It is the same as gauge pressure.
   b. It is equal to the atmospheric pressure.
   c. It is atmospheric pressure plus the gauge pressure.
   d. It is the pressure condition of being at 100% vacuum.

128. Which of the following describes saturation temperature?
   a. It is the temperature at which the refrigerant boils, but is independent from pressure.
   b. It is the temperature at which the refrigerant boils, but varies with pressure.
   c. It is the same as critical point for the gas refrigerant.
   d. It is the point where the ice starts to melt.

129. Which of the following modern refrigerants is the most preferred refrigerant because of non-toxicity and being environment-friendly?
   a. R-22
   b. R-134a
   c. R-404A
   d. R -

130. Which of the following does the term “latent heat” describe when referring to a refrigerant?
   a. The heat required to cause evaporation at constant boiling temperature and pressure.
   b. The heat required to raise the liquid temperature to boiling point.
   c. The heat required to superheat the refrigerant gas.
   d. The temperature.

(Topic No. 41)

131. The ship’s main engine is used to turn It's __________ and move the ship through the water.(revise question)
   a. compressor
   b. propeller
   c. rudder
   d. pump

132. What kind of engine prime mover is used onboard ships for its main propulsion that typically requires high-voltage power?
   a. internal combustion engines
   b. steam turbines
   c. electric motors
   d. gas engines

(Topic No. 42)

133. Which engine-room auxiliary machinery is used to heat up water to evaporate and generate and accumulate steam needed in engine room operations?
   a. Purifier
   b. Hydrophore
   c. Auxiliary boiler
   d. Freshwater generator

134. Which engine-room auxiliary machinery is used to get rid of the unwanted contaminants and waterants from the lube oil?
   a. purifier
   b. hydrophore
   c. auxiliary boiler
   d. freshwater generator

(Topic No. 43)

135. Which steering gear drive machinery uses a hydraulic rotary actuator to change the position of the rudder?
   a. Ram drives
   b. Vane drives
   c. Cam drives
   d. None of the above

136. Which steering gear drive machinery uses a pair of hydraulic linear actuator to change the position of the rudder?
   a. Ram drives
   b. Vane drives
   c. Cam drives
   d. None of the above

(Topic No. 44)

137. Which of the following deck machinery equipment is used to lift heavy cargo loads?
   a. crane (revised or changed question)
   b. pump
   c. winch
   d. compressor

(Topic No. 45)

138. Which of the following deck machinery equipment is used to reel in, to let out, or otherwise adjust the tension of a rope, wire, or cable? changed question)
   a. crane
   b. pump
   c. winch
   d. compressor

(Topic No. 46)

139. Which of the following hotel equipment is essential for quick-drying washed clothes, using heaters and electric motor spinners.(question)
   a. Air Conditioner
   b. Tumble Dryer
   c. Compressed-air Dryer
   d. Blower

(Topic No. 47)

140. What kind of interlock system is used for the main engine that is located in the engine flywheel?
   a. Reversing interlock
   b. Turning gear interlock
   c. Auxiliary blower interlock
   d. Air spring pressure interlock

141. Diesel engines are started by supplying compressed air into the cylinders in the appropriate sequence for the required direction, these sources are stored in containers called
   a. air panels
   b. air valves
   c. air blocks
   d. air bottles

(Topic No. 48)
142. Which of the following best describes what we mean by “power factor” in an AC electrical system?
   a. It is the proportion of apparent power which is converted to reactive power.
   b. It is the ratio of power drawn from the generator to its maximum capacity.
   c. It is the proportion of apparent power which is delivered as reactive power.
   d. It is the ratio of real power and apparent power.

143. Which of the following best describes why traditionally ships’ electrical systems have operated with an insulated neutral?
   a. To minimize the risk of generator damage.
   b. To minimize the cost of the electrical network
   c. To minimize the risk of essential equipment tripping.
   d. To minimize the risk of electric shock or electrocution of personnel.

144. How many electrical degrees are there between the phases in a 3-phase AC generator?
   a. 60 degrees
   b. 90 degrees
   c. 120 degrees
   d. 180 degrees

145. Which of the following electrical parameters is not a requirement in the synchronization of generators?
   a. Voltage
   b. Current
   c. Frequency
   d. Phase Sequence

146. What does it mean when the synchronizing lamp rotation is counter-clockwise?
   a. The frequency of the incoming generator is too slow in reference to that of the main bus bars.
   b. The frequency of the incoming generator is correct, but the voltage amplitude is still incorrect.
   c. The frequency of the incoming generator is too fast in reference to that of the main bus bars.
   d. The frequency of the incoming generator and the voltage amplitude are both incorrect.

147. The power factor of an AC generator depends on the ___________.(revised choices)
   a. impedance load.
   b. core losses.
   c. armature losses.
   d. speed of the rotor.

148. What will happen if the load dependent start/stop operation of generators is in “Auto Mode”?
   a. It will start/stop the generator sets in response to varying load.
   b. It will start/stop the generator sets in response to fuel consumption.
   c. It will start but never stop the generator sets in response to varying load.
   d. It will stop but never start the generator sets in response to varying load.

149. Which of the following best describes an over-excited alternator?
   a. Generator is operating at unity power factor.
   b. Generator is operating at lagging power factor.
   c. Generator is operating at leading power factor.
   d. Generator is operating at lagging to leading power factor.

(Topic No. 49)
150. Which of the following is the typical instrumentation signal range for monitoring systems on ships?(changed)
   a. 0 mV – 20 mV

b. 0 V – 20 V
c. 4 mA – 20 mA
d. 4 A – 20 A

151. Where will you find the instruments for synchronizing the generators?(changed)
a. On the generator
b. On the diesel engine
c. On the circuit breaker
d. On the main switchboard

152. Which one of the following best describes the meaning of "discrimination" when applied to electrical distribution systems onboard ships?
a. Color coding is used to "discriminate" between conductors made with flame retardancy and those without.
b. Color coding is used to "discriminate" between conductors made with PVC and those made of other materials.
c. Protection devices are to "discriminate" between healthy circuits and circuits with faults, thereby shutting off only the faulty circuits.
d. Protection devices are to "discriminate" between high-voltage circuits and low-voltage circuits, shutting off only the low-voltage circuits.

(Topic No. 53)
153. Which of the following best describes a marine high voltage equipment?(Choices)
a. Rated at 440 V AC and above
b. Rated at voltages above 690 V AC
c. Rate at 1000 V AC and above
d. Rated at voltages above 1000 V AC

154. Ships fitted with high-voltage installations tend to use earthed neutral electrical networks. Which one of the following best describes how excessive fault currents are prevented in the event of an earth fault on these ships?
a. An Impedance is placed in neutral earthing line.
b. Earth leakage circuit breakers are used on every circuit.
c. The risk of earth fault is reduced by extra planned maintenance.
d. An Impedance is placed in each output cable from the generators.

155. Which of the following high-voltage testing tools are necessary to test the arc suppression medium in contact cylinders of high-voltage vacuum circuit breakers and high-voltage vacuum contactors?
a. HV detectors
b. Integrity testers
c. Pressure testers
d. HV insulation testers

156. Which one of the following best describes why it is normal practice to place the high-voltage tester in its test unit after a circuit has been tested as being dead?
a. To make sure it is still working correctly
b. To protect the ceramic end of the teste.
c. To make sure that the test unit batteries are still working
d. To make sure that the test unit reading agrees with the circuit reading

(Topic No. 36)
157. Which one of the following are normally fed into the ECDIS system?
a. The ship's internal communication system via the LAN
b. GMDSS communication system from the radio equipment
c. Ship's stability information
d. Navigational signal

158. A bridge navigation equipment that can detect targets and display the information on the screen such as the distance of the ship from land, any floating objects other vessels, and obstacles to avoid a collision.
a. RADAR system

b. GMDSS system
c. Gyro compass
d. Auto Pilot

(Topic No. 37)

159. What is the function of a squelch in radio communication equipment?
   a. Dual watch
   b. Automatic gain control
   c. Eliminates audio noise when there is no traffic
   d. Automatic tuning

160. Which of the following refers to the digital communication equipment used to connect the computers in the bridge to that of the computers in the engine control room?
   a. Telephone
   b. Ethernet Switch
   c. Branch Exchange
   d. Internet Router

(Topic No. 38)

161. What is the most important factor for the range of a VHF station?
   a. The output power
   b. The power supply
   c. The size of the antenna
   d. The height of the antenna

162. On which VHF DSC channel shall all GMDSS equipped ships keep watch?
   a. Ch. 70
   b. Ch. 16
   c. Ch. 13
   d. 2187.5 kHz

(Topic No. 39)

163. In working onboard with electrical and electronic systems operating in flammable areas, which of the following safety marking is essential to look for?
   a. ANSI
   b. EX
   c. IEC
   d. NEMA

164. What electronic circuit or component is used in intrinsic safety requirements in order to limit the voltage and current into a hazardous flammable area onboard ship?
   a. Diode Rectifiers
   b. Zener Barriers
   c. Capacitor Filters
   d. Circuit Clampers

(Topic No. 6)

165. Which of the following codes or books was established to set the minimum international requirements on training, certification, and watchkeeping for seafarers based on the capacity they will be serving onboard a particular type of vessel?
   a. STCW
   b. ISM
   c. IMDG
   d. SMCP

166. A seafarer’s main document, received upon completion of the following – a training program approved by the issuing administration, an approved period of approved seagoing service, and the passing of assessment of competence based on the level of responsibility and onboard capacity.
   a. PRC
   b. PEC
   c. COP
   d. COC

(Topic No. 7)

167. Which management technique may help personnel prioritize tasks more efficiently, improve the balance of work across the team, and produce quality outputs?
   a. Resource management
   b. Training management
   c. Workload management
   d. Time management

168. In shipboard operations, the use of maritime English and each compartments like establishing an English language only in the workplace as a policy ensures which of the following awareness among team members.
   a. Situational awareness
   b. Safety awareness
   c. Environmental awareness
   d. Cultural awareness

(Topic No. 8)

169. It is a process of planning, scheduling, and allocating people, money, and equipment to a task onboard to achieve a desired output?
   a. Resource management
   b. Training management
   c. Workload management
   d. Time management

170. In effective resource management, this refers to the understanding of how personnel acquire their cultures, personal identities, life ways, and mental and physical health of individuals and communities.
   a. Situational awareness
   b. Safety awareness
   c. Environmental awareness
   d. Cultural awareness

Part V. **Applying electro-technical engineering principles.** Use a separate sheet of paper for your answer (short answer).

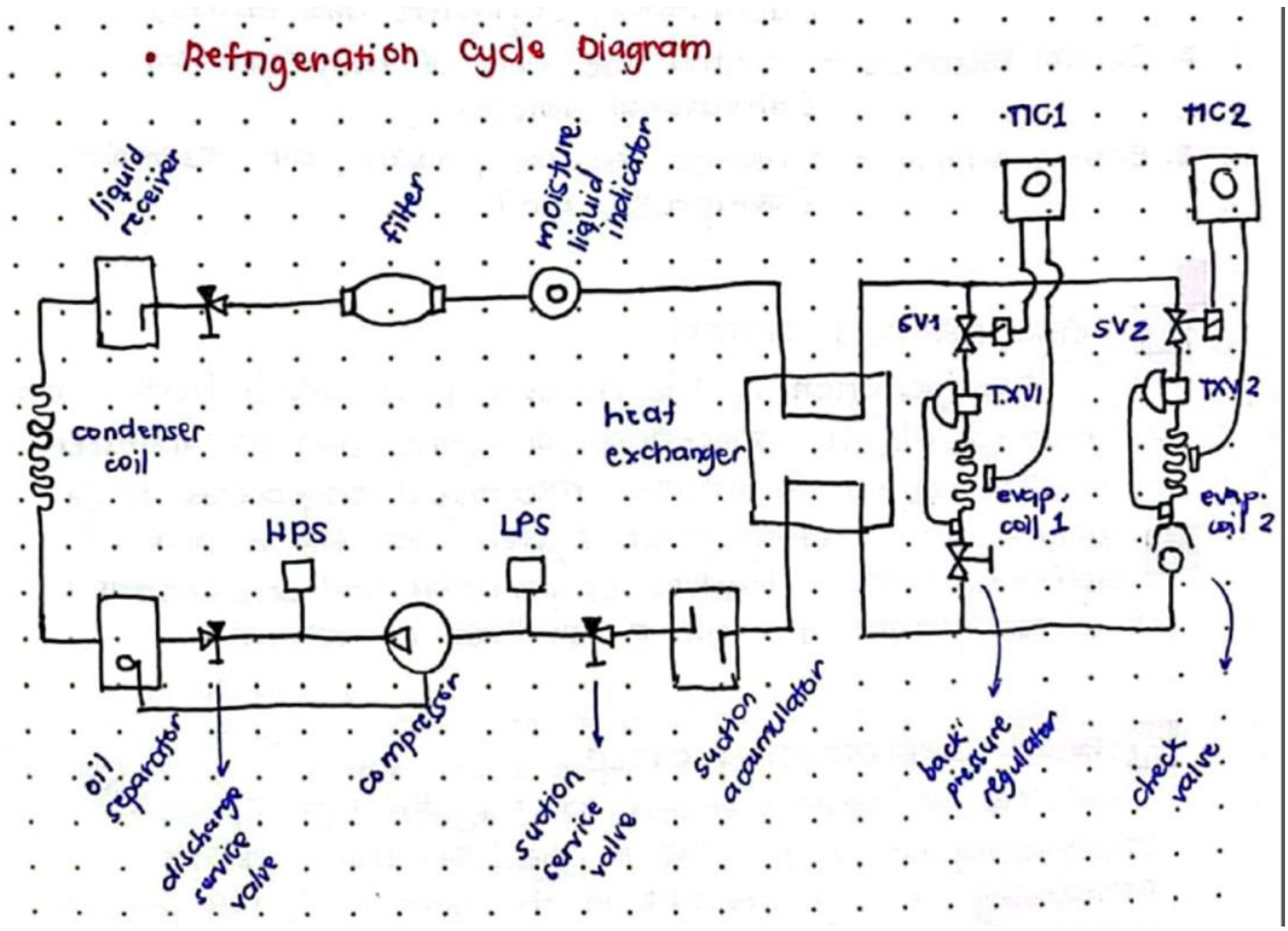
(Topic No. 10)

171. Enumerate the different heat transfer methods.
   - Convection
   - Conduction
   - Radiation

172. Compare each of the types of heat transfer method from each other for shipboard application
   Convection – movement through molecules (boiler)
   Conduction – physical connection (sensors)
   Radiation – electromagnetic waves (microwave)

(Topic No. 40)

173. Draw 4 basic components a schematic diagram for a typical marine refrigeration system. Appropriately label each component and properly indicate the flow of the refrigerant in the process.



174. Explain the operation of the marine refrigeration equipment from the schematic diagram that you have created.
- Refrigerant is drawn to the compressor to be compressed into a high pressure, high temperature vapor. The compressor discharges it to the condenser to give up heat, condensing vapor to liquid then passes through the thermal expansion device where it reduces the pressure and temperature. The refrigerant that exits the valve becomes low pressure and low temperature liquid. It then enters the evaporator to gain heat and evaporates liquid turning it to low pressure vapor. Then starts another cycle.

(Topic No. 11)

175. Differentiate energy from power by comparing its units of measurement in si unit
Energy – Joules
Power - Watts (joules/second)

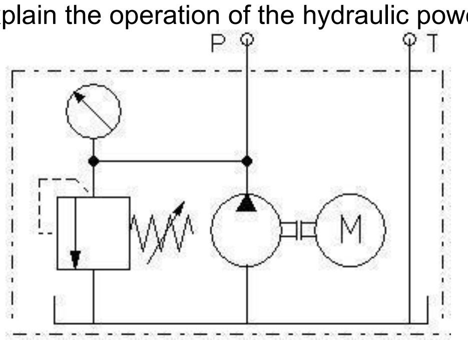
176. Explain how work is done in the conversion of electrical energy into mechanical energy for electric motors.

Induction motors are efficient, simple, and reliable at converting electrical energy into mechanical motion. The stator, the stationary outer part of the motor, houses electromagnets arranged in coils. When AC current flows, it creates a rotating magnetic field. The rotating magnetic field from the stator cuts through the conductive bars or windings in the rotor, the inner part of the motor. This current creates its own magnetic field, interacting with the stator field. This interaction creates a force that turns the rotor (torque) The rotor keeps spinning as the magnetic dance continues.

(Topic No. 12)

177. Discuss the relationship between pressure and force in a fluid system.in si unit
Force – Newton
Pressure – Pascal, Bar, Psi, mmHg

178. Explain the operation of the hydraulic power pack below. (replace drawing with return filter)



- MAIN PARTS
  - Pump - operates on the use of pressurized gas to move media.
  - Drive Motor - run a compressor, making compressed air
  - Tank / Reservoir - its primary function is to act as temporary storage to accommodate peak demands of compressed air.
- ADDITIONAL PARTS.
  - Pressure Relief Valve - implemented to prevent spikes in pressure
  - Manometer - a device that is able to measure the pressure of a medium
  - Accumulator - energy storage, shock absorption, pressure regulation
  - Filters - remove contaminants (water, oil vapour) from a compressed airstream.

(Topic No. 14)

179. Discuss what happens to the resistance of a pure metal when it is experiencing an increase in temperature?

THE RELATIONSHIP OF PURE METAL TO TEMPERATURE

The relationship between pure metals and temperature is governed by fundamental physical principles.

- Melting Point - higher purity leads to a higher melting pt.
  - Iron (1538 °C)
  - Alloy of Iron and Carbon (1450 °c)
- Electrical Conductivity
  - Copper
  - Silver
- Thermal Expansion
  - Aluminum
- Strength

CS CamScanner

In addition, to the factors listed, there are other factors that can affect the relationship between pure metals and temperature.

- the presence of impurities
- the crystal structure of the metal
- the amount of stress on the metal

180. Describe the physical appearance of a spring when it is stretched for a long period of time.

SPRING WHEN SUBJECTED TO WEIGHT FOR A LONG TIME

When a spring is subjected to a constant weight for an extended duration, several physical properties are affected:

DEFLECTION – the spring will initially deflect or compress under the weight.

STIFFNESS – resistance to deformation.

YIELD STRENGTH – maximum stress the spring can withstand; if the applied weight exceeds the yield strength, the spring may take a permanent set when the weight is removed.

FATIGUE – if repeated loading and unloading, it can lead to cracks and eventual failure.

(Topic No. 6)

181. What is the main job of an Electro-Technical Officer onboard a ship? Explain role by elaborating the tasks that an ETO will need to perform onboard the ship.

DUTIES OF AN ETO

- Regularly checks and inspects all electro-technical equipment, motors, generators, switchboards, and navigation systems.
- Performs scheduled maintenance tasks as per manufacturer's recommendations and company guidelines.
- Troubleshoots and repairs electrical and electronic faults, ensuring timely solutions to minimize downtime.
- Coordinates with onboard engineers to identify and resolve electrical issues.
- Monitors and controls the ship's power generation and distribution systems.
- Operates navigation and communication systems
- Maintains emergency power systems and ensure readiness.
- Complies with all maritime regulations and safety procedures.
- Maintains accurate records of all maintenance work, inspections, and repairs

(Topic No. 37)

182. Enumerate at least 3 internal communications equipment that can be found onboard a ship, briefly explain the purpose of each equipment.

Communication Systems

INTERNAL

- **Automatic Telephone System** - the energy comes from the main supply source and emergency supply. It is primary or main internal communication system use onboard to communicate from different fixed location of the ship
- **Battery-less Telephone System** - used for communicating with the wheelhouse, engine control room, and steering gear room in case of emergency
- **Talkback System** - offers communication between the muster stations and a number of sub stations located in different parts of the vessel where communication is needed
- **Public Address System** - combination of a set of audio equipment that allows to broadcast over a designated area or to the entire vessel

(Topic No. 38)

183. Enumerate at least 3 external communications equipment that can be found onboard a ship, briefly explain the purpose of each equipment.

EXTERNAL

Global Maritime Distress and Safety System (GMDSS) - to guarantee that complying vessels will be able to communicate with an onshore station at any time, from any location, in case of distress or to exchange safety-related information

GMDSS Equipment:

- **VHF with DSC**
- **Search and Rescue Transponder (SART)**
- **NAVTEX receiver**
- **Enhanced Group Call (EGC) receiver**
- **VHF portable**
- **MF/HF DSC**
- **Emergency Position Indicating Radio Beacon (EPIRB)**
- **Inmarsat-B or Inmarsat-C**
- **MF/HF telephony with DSC and telex**

SART – a specialized device used primarily in locating vessels, aircraft or individuals in distress during search rescue operation

Part VI. **Interpreting Schematic Diagrams.** Refer to this diagram and use a separate sheet of paper for your answers.

(Topic No. 21)

184. In the given electrical equipment schematic diagram below, encircle and label what is the power circuit and what is the control circuit.

185. Using the given schematic diagram, explain where a disconnection and isolation procedure should be done to ensure safety prior to any electrical maintenance works.

(Topic No. 23)

186. In the given schematic diagram above, explain the operation of the circuit.

187. Distinguish the difference between the protection provided by the overload relay (OL) and the circuit breaker (CB) in the electrical control system. Use a separate sheet of paper for your answer.
Overload Relay – Motor protection
- Overload only
- delayed response time
- single device

Circuit Breaker – Circuit protection
- Overload and short circuits
- Instant or delayed response time
- Entire circuit

(Topic No. 48)

188. For the equipment in the given circuit diagram, a problem malfunction was reported for the pump equipment. It was reported that during the duty engineer's initial investigation they have identified that the equipment was working only on slow operation after pressing the START pushbutton for supplying air but was not delivering enough pressure into the system. What will your initial analysis be with the report and the function test conducted by the duty engineer?
- there may be a blockage or restriction in the air intake, filters or piping could reduce air flow
- pump is worn or damaged, it might not generate adequate pressure despite running
- insufficient voltage or current

189. Create a step plan to describe the troubleshooting that you will be doing for the said scenario above.

(Topic No. 9)

190. Use the attached UMTC Toolbox Risk Identification Permit provided for the troubleshooting activity that you have planned for.

Part VII. **Troubleshooting**. Use a separate sheet of paper for your answers.

(Topic No. 31)

191. A pt100 alarm indicates a high temperature reading in the Cooling Water Temperature HT Engine Outlet, but eventually verified that the local reading is still normal at 85oC using a thermometer gun. Identify the possible causes of the error and explain why. Refer to the Reference Tables provided.
- PT100 is already defective
- PT100 is not properly calibrated

192. The control system for the Exhaust Gas Temperature after the Cylinders of the Main Engine has the following specifications:

Sensor Type:	Thermocouple
Normal Values:	less than 500^{o}C
Sensor-Range:	0 … 1000oC
Signal-Range:	0 … 10 V

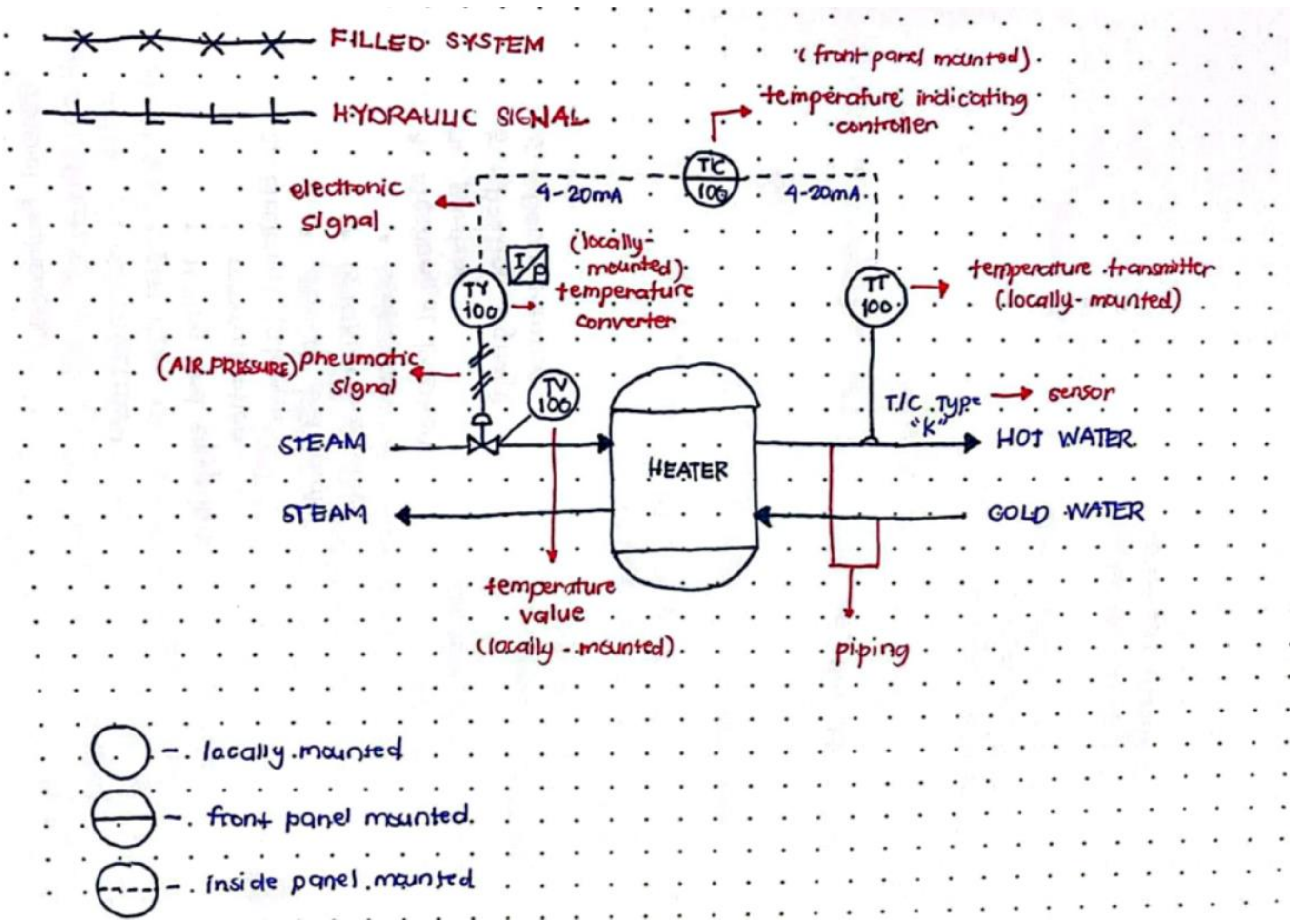
In normal operation the system reading is 450oC. When a low temperature indication of 100oC was read in the control room monitor, but eventually verified locally as normal at 450oC using an infrared thermometer, explain the possible causes of the inconsistency and explain why. Refer to the Reference Tables provided.

(Topic No. 28)

193. Draw an equivalent piping and instrumentation diagram (P&ID) for the given process control loop described by the illustration below. [1] Use appropriate ISA standard symbols and instrumentation tags – Refer to the Reference Tables provided. [2] Properly label each stage of the process control loop by indicating signal range values.

CONVERTER

PID CONTROLLER



Part VIII. **Marine High Voltage Practice.** Use a separate sheet of paper for your answer.

(Topic No. 50)

194. Enumerate the 3 main electrical high-voltage hazards that can be encountered onboard a ship with high-voltage installations, and briefly explain each to compare with one another.

3 HIGH VOLTAGE HAZARDS

1.) Electric Shock — This is the most immediate and potentially fatal risk associated with high voltage. Direct contact with live wires or equipment can cause severe burns, muscle contractions, and death.

2.) Arc Flash and Arc Blast — when high voltage arcs across an air gap, it creates an intense flash of light, heat, and pressure. It causes severe burns and can ignite nearby flammable materials. The arc blast, a powerful shockwave from the expanding hot air can cause injuries and damage equipment.

3.) Fires and Explosions — high voltage faults or equipment malfunctions can cause fires and explosions, leading to widespread damage and potential loss of life.

195. Identify important marine procedures and safe practices in addressing these high-voltage safety concerns.

3 CRUCIAL SAFETY PROCEDURES FOR HV OPERATIONS ONBOARD SHIP

1.) Permit to Work System

* Before any HV work begins, a formal permit to work system must be implemented. This ensures proper planning, risk assessment, and authorization.

- work scope.
- potential hazards
- required safety precautions
- personnel involved
- emergency procedures.
- authorized and qualified personnel

2.) LOTOTO (Lock-out, Tag-out, Try out) Procedures

* Isolating and securing high voltage equipment before intervention is paramount.

* These procedures ensure de-energizing the equipment, locking out isolation points, and applying visible tags indicating "DO NOT OPERATE" or similar warnings.

3.) PPE (Personal Protective Equipment) and Safe Work Practices

* Wearing appropriate PPE is essential to minimize the risk of electric shock, arc flash etc.

- flame-retardant and insulated coveralls
- gloves, boots, helmet.
- arc flash face shield.
- safety glasses
- rated tools & equipment.

* Maintain safe working distance.

(Topic No. 52)

196. Explain the difference between an insulated neutral system and a grounded neutral system for ship electrical power distribution systems.

INSULATED VS. EARTHED NEUTRAL

- Insulated Neutral

  The neutral wire is completely isolated from the ship's hull (earth). It is safer because in case of a single earth fault (leakage), current does not flow through the hull, reducing shock and fire risk. It is also less prone to accidental grounding. Although it is safer, fault detection can be challenging.

- Earthed Neutral

  The neutral wire is directly connected to the Ship's hull, it is easier for fault detection as it triggers safety devices if leakage current flows, It is considered having increased risk during faults and ground faults can cause unnecessary tripping.

(Topic No. 53)

197. What are the 3 types of arrangement for a vessel to be classified as a ship with marine high-voltage installation?
   - generation
   - distribution
   - utilization

198. Discuss the operation of the general high-voltage safety procedure of Disconnection, Isolation, and Earthing.

"DIE" ON HV ELECTRICAL SYSTEMS

D.I.E. stands for Disconnect, Isolate, and Earth. It is a crucial safety procedure employed before any work is done on HV electrical systems.

"Disconnect"

Physically disconnecting the equipment from the high voltage source. This involves opening circuit breakers, removing fuses etc. this ensures that no live current can flow.

"Isolate"

After disconnecting, additional information for insulation measures are implemented. This often involves physically locking out isolation points to prevent accidental reconnection.

"Earth"

Once disconnected and isolated, the equipment is earthed by connecting it to the ship's hull grounding system. This provides a safe path for any fault currents to discharge.

During normal inspection of high voltage switchboard, a thermal imaging camera has indicated a high temperature region at a busbar section of the bow thruster motor that will require investigation or possible repair. In order to carry out the investigation, the high voltage switchgear of the bow thruster must be isolated and made safe.

(Topic No. 53)

199. List down required PPE and their classifications and category markings.

PPE FOR HIGH VOLTAGE SYSTEMS

- Flame-retardant and insulated coveralls
- Arc Flash face shield and safety glasses
- Electrical safety-rated gloves and boots
- Hard hat
- Hearing Protection.

In Voltage Levels:

High voltage:

1kV AC to 36kV AC / 1.5kV DC to 50kV DC — IEC 60079-11 category 3 or 4

above 36kVAC or 50kVDC — IEC 60079-11 category 4 or higher

Choose PPE with an ATPV (Arc thermal Performance Value).

200. Write down the steps for isolating the bow thruster switchgear and motor to be submitted to the Senior Officer for approval.

ISOLATION PLAN OF BOW THRUSTERS

Information Required:

- Ship's details
- Bow thruster details — model, power rating, manufacturer.
- Isolation Procedures
- Safety regulations

P&ID – PIPING AND INSTRUMENTATION DIAGRAM

	Instrumentation Signal type	Power Supply/ Source	Minimum 0%	Maximum 100%
ELECTRICAL	A. CURRENT	24VDC	4mA	20mA
	B. VOLTAGE	24VDC	1 or 0V	5 or 10V
PRESSURE	A PRESSURE (SI)	1-4 BAR	0.2 BAR	1.0 BAR
	B PRESSURE (ENG)	20 PSI	8 PSI	